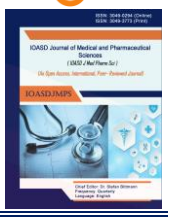
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Influence of Family Involvement in the Treatment of Patients with Mental Illness in the University of Port Harcourt Teaching Hospital (UPTH): A Descriptive Cross-Sectional Assessment of Extent, Outcomes and Barriers

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ABSTRACT

Background: Mental illness remains a significant public health concern in Nigeria, where limited resources and inadequate community support make family involvement an important component of patient recovery. However, empirical evidence on family participation in tertiary mental healthcare remains limited. ***Aim:*** This study assessed the influence of family involvement on the treatment of patients with mental illness at the University of Port Harcourt Teaching Hospital (UPTH), focusing on its extent, relationship with treatment adherence, relapse and recovery, and barriers to effective involvement. ***Method:*** A descriptive cross-sectional design was adopted. Data were collected from 135 primary family caregivers using a structured, self-administered questionnaire validated by experts and pilot-tested, with a Cronbach's alpha of 0.81. Purposive sampling was used. Data were analysed using frequencies, percentages, means, standard deviations and multiple linear regression at a 0.05 significance level. ***Results:*** Family involvement was high overall (mean = 4.43, SD = 0.67), particularly through financial support (mean = 4.84), emotional support (mean = 4.75) and medication monitoring (mean = 4.69), while involvement in clinical decision-making was lower (mean = 3.51). Caregivers strongly agreed that family involvement improved treatment adherence (mean = 4.67) and reduced relapse while promoting recovery (mean = 4.63). Financial constraints (mean = 4.87), caregiver burden (mean = 4.71), and stigma (mean = 4.38) were major barriers. Regression analysis demonstrated a significant positive relationship between family involvement and treatment adherence ($R = 0.803$, $R^2 = 0.645$, $F = 78.41$, $p < 0.001$). ***Conclusion:*** Family involvement at UPTH is high and significantly associated with improved treatment adherence and recovery outcomes, despite substantial financial and psychosocial challenges. Strengthening nurse-led family psychoeducation, reducing financial barriers and addressing stigma are recommended to promote effective family-centred mental healthcare.

Keywords: Family involvement, mental illness, treatment outcome, patient recovery, treatment adherence, relapse, caregiver burden, University of Port Harcourt Teaching Hospital.

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1. INTRODUCTION

Mental illness remains a significant public health concern in Nigeria, yet the role of family involvement in patient treatment is often underutilized. Mental health services in Nigeria are heavily burdened, with limited resources and inadequate community support structures, making family participation a potentially critical component in patient recovery (Gureje *et al*., 2020). The neglect or absence of family support can delay recovery, increase relapse rates, and reduce treatment adherence among mentally ill patients. Given the resource-limited setting of the University of Port Harcourt Teaching Hospital (UPTH), understanding

how family participation impacts the treatment journey is central to improving mental health outcomes, and calls for a deeper examination of how the presence or absence of family involvement influences patient care.

The treatment outcome of patients with mental illness includes medication adherence, relapse rate, symptom reduction, and the overall recovery trajectory of the patient. Patients with strong family support are more likely to adhere to treatment regimens and exhibit better recovery rates than those without such support (Ogunsemi *et al*., 2021), highlighting the importance of measuring treatment success not only through clinical interventions but also through social and familial dynamics. The level and type of family involvement in the treatment process includes attending therapy sessions, supervising medication intake, offering emotional support, and maintaining communication with healthcare providers. Structured family involvement has been shown to reduce hospital readmissions and improve functional outcomes among individuals with mental illness (Ebigbo & Bakare, 2019), shifting the treatment model from being purely hospital-based to a more collaborative, person-centred approach.

Empirical studies have consistently established a positive relationship between family involvement and improved mental health treatment outcomes. For example, patients receiving coordinated care with family participation were found to be 40% less likely to relapse than those receiving care without family input (Abiodun *et al*., 2022), suggesting that the family serves as both a social support system and an informal extension of the healthcare team, especially in resource-limited settings. Given the strain on mental health professionals in UPTH, involving family members could fill critical gaps in psychosocial support and daily supervision, reinforcing the importance of systematically assessing the role of family in treatment planning and implementation. Strengthening family involvement not only improves patient care but also aids in demystifying mental health conditions within society (Oyekunle & Nwankwo, 2021), so this study has practical implications for mental health policy, nursing practice, and family education initiatives.

Mental illness presents a significant burden in Nigeria, with limited access to mental health services and widespread social stigma contributing to poor treatment outcomes. Despite efforts by healthcare providers, a gap remains in the provision of comprehensive care, especially in tertiary institutions like UPTH. One of the often-overlooked components of effective mental healthcare is the role of the family in the treatment process. In many cases, patients are either abandoned, stigmatized, or inadequately supported by their families, which leads to frequent relapses, non-adherence to treatment, and long-term institutionalization. Although evidence from other countries and some parts of Nigeria suggests that family involvement can improve patient outcomes, there is limited empirical research focusing on this within the UPTH context, and the absence of clear protocols on family engagement in psychiatric care raises concerns about missed opportunities for holistic healing.

This study therefore examines the influence of family involvement in the treatment of mentally ill patients at UPTH, exploring how different forms of family engagement impact treatment adherence, recovery outcomes, and the barriers to sustained participation, in order to provide evidence-based recommendations for integrating family-oriented strategies into mental health care delivery in UPTH.

2. MATERIALS AND METHOD

2.1 Study Design

This study adopted a descriptive cross-sectional research design, appropriate for examining the current perceptions and experiences of family caregivers regarding their involvement in the treatment of mental illness. A descriptive design allows for the observation and documentation of variables as they exist naturally, without manipulation, and was suitable for understanding how family involvement affects the treatment process and outcomes among mental health patients in a real-life setting such as UPTH.

2.2 Study Setting

The study was conducted at the University of Port Harcourt Teaching Hospital (UPTH), located in Port Harcourt, Rivers State, Nigeria. UPTH is a tertiary healthcare institution that serves as a referral centre for patients within Rivers State and neighbouring states. The hospital has a functioning Psychiatric/Mental Health Unit that provides psychiatric evaluation, psychotherapy, medication management, inpatient care, and community-based outreach programmes, and was chosen for this study due to its diverse patient population and its central role in mental health service delivery in the South-South region of Nigeria.

2.3 Participants and Sampling

The target population comprised adult patients diagnosed with mental illness who were receiving treatment at UPTH and their family members actively involved in their care. The accessible population comprised patients attending the psychiatric clinic and admitted inpatients within the psychiatric ward, alongside one key family caregiver per patient. The sample size was calculated using Taro Yamane's formula at a 5% margin of error, yielding a sample of 135 primary family caregivers, and the study used a purposive sampling technique due to the limited accessibility to inpatient mental health patients and the sensitive nature of psychiatric care. Inclusion criteria comprised family members of patients diagnosed with a mental illness for over five years and currently receiving treatment at UPTH, who were directly involved in the patient's care (e.g., attending clinic visits, providing home care, or

emotional support) and who gave informed consent. Family members of patients who were acutely psychotic or mentally unstable, family members who were not actively involved in treatment, and individuals who declined participation or withdrew during the study were excluded.

2.4 Data Collection Instruments

The primary instrument for data collection was a structured, self-administered questionnaire developed by the researcher, divided into four sections: Section A, demographic information; Section B, the nature and extent of family involvement; Section C, perceived impact of family involvement on treatment outcomes; and Section D, challenges and barriers to effective family involvement. Data were collected through face-to-face distribution of questionnaires to eligible participants; the researcher explained the purpose of the study, obtained informed consent, and guided respondents as needed, providing assistance while maintaining confidentiality where patients were unable to complete the questionnaire independently.

2.5 Validity and Reliability

To ensure validity, the questionnaire was subjected to expert review by professionals in psychiatric nursing, public health, and clinical psychology, whose feedback helped refine the items for clarity, relevance, and content coverage. The instrument's reliability was assessed using a pilot study involving 20 family members at a similar psychiatric unit outside the study setting, which produced a Cronbach's alpha coefficient of 0.81, indicating a high level of internal consistency.

2.6 Ethical Considerations

The study adhered strictly to ethical guidelines for research involving human subjects. Approval was sought from the Ethical Review Committee of the University of Port Harcourt Teaching Hospital (UPTH). Participants were fully informed about the nature and purpose of the research, their right to decline or withdraw at any stage, and the confidentiality of their responses.

3. Data Analysis

Data obtained were analysed using descriptive statistics such as frequencies, percentages, and mean scores, with results presented in tables for clarity. To test the study's hypothesis on the relationship between family involvement and treatment outcomes, Multiple Linear Regression analysis was used at a 0.05 significance level, applying the decision rule that the null hypothesis was rejected where $p < 0.05$ and retained where $p \geq 0.05$.

4. RESULTS

A total of 135 primary family caregivers, drawn from the accessible population at the Psychiatric/Mental Health Unit of UPTH, completed and returned the questionnaire, representing the full analysed sample (N = 135).

4.1 Socio-Demographic Characteristics

The majority of primary caregivers were female (64.4%), confirming that the burden of primary care for mental illness at UPTH largely rests on female family members (Table 1). The largest proportion of respondents fell within the 30–39 years age bracket (35.6%), indicating that caregivers are typically young to middle-aged adults balancing caregiving responsibilities with career and family demands, and a significant majority were married (62.2%). The educational profile of respondents was high, with the largest group possessing a B.Sc./B.A./B.Ed (44.4%), suggesting a population with strong potential for health literacy. The most common relationship to the patient was spouse (44.4%), followed by parent (24.4%), emphasising the critical involvement of immediate nuclear family members in the care process.

Table 1: Frequency and Percentage Distribution of Respondents by Demographic Variables (n = 135)

Variable	Category	n	%
Gender	Male	48	35.6
	Female	87	64.4
Age Group (years)	20–29	18	13.3
	30–39	48	35.6
	40–49	39	28.9
	50 & above	30	22.2
Marital Status	Single	33	24.4
	Married	84	62.2
	Separated/Divorced	12	8.9
	Widowed	6	4.4
Educational Level	Secondary/WASC	15	11.1
	OND/NCE/HND	36	26.7
	B.Sc./B.A./B.Ed	60	44.4
	Postgraduate	24	17.8
Relationship to Patient	Spouse	60	44.4
	Parent	33	24.4
	Sibling	24	17.8

Variable	Category	n	%
	Other Relative	18	13.3

Source: Field Survey Data, UPTH, 2025.

4.2 Extent of Family Involvement in Patient Treatment

Research Question 1 sought to determine the level of family involvement using a 5-point Likert scale (1 = Very Low Extent to 5 = Very High Extent), with a criterion mean of 3.50 or above indicating a high extent of involvement. The overall mean score of 4.43 (Table 2) indicates a High Extent of family involvement in the treatment of patients with mental illness in UPTH. All items, with one exception, scored within the 'High' or 'Very High' extent categories. The item with the highest mean score was 'Provide financial support for the patient's treatment' (mean = 4.84), followed closely by 'Offer emotional support during recovery periods' (mean = 4.75) and 'Monitor medication administration and adherence' (mean = 4.69), suggesting that involvement is highest in areas requiring direct, logistical, and affective support. The lowest mean score was recorded for 'Engage in decisions regarding the patient's treatment plan' (mean = 3.51); although still classified as 'Moderate', this is substantially lower than all other dimensions, suggesting a structural gap where family members, despite high levels of practical involvement, perceive a lesser role in formal clinical decision-making.

Table 2: Extent of Family Involvement in Patient Treatment (n = 135)

Item	Mean	SD	Extent of Involvement
Monitor medication administration and adherence.	4.69	0.57	Very High
Attend therapy or counseling sessions with the patient.	4.38	0.79	High
Provide financial support for the patient's treatment.	4.84	0.36	Very High
Offer emotional support during recovery periods.	4.75	0.53	Very High
Engage in decisions regarding the patient's treatment plan.	3.51	1.12	Moderate
Overall Mean & SD	4.43	0.67	High Extent

Source: Field Survey Data, UPTH, 2025.

4.3 Influence of Family Involvement on Treatment Adherence

Research Question 2 investigated caregivers' perceptions of the influence of their involvement on patient treatment adherence, using a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). The overall mean score of 4.67 (Table 3) indicates a definitive and strong agreement that family involvement positively influences patient treatment adherence. The item with the highest mean score, 'My emotional support has a positive impact on the patient's willingness to follow care instructions' (mean = 4.89), suggests that affective factors are the most powerful components of adherence, while high scores were also recorded for logistical support in compliance, such as adherence to medication schedules (mean = 4.71) and appointment attendance (mean = 4.67).

Table 3: Influence of Family Involvement on Patient Treatment Adherence (n = 135)

Item	Mean	SD	Level of Agreement
My involvement helps the patient adhere to medication schedules.	4.71	0.51	Strongly Agree
The patient is more likely to attend appointments when I am involved.	4.67	0.55	Strongly Agree
The patient recovers faster when I participate in therapy/counseling.	4.40	0.82	Agree
My emotional support has a positive impact on the patient's willingness to follow care instructions.	4.89	0.31	Strongly Agree
Overall Mean & SD	4.67	0.55	Strongly Agree

Source: Field Survey Data, UPTH, 2025.

4.4 Influence of Family Involvement on Relapse Rate and Overall Recovery Trajectory

Research Question 3 assessed the influence of family involvement on relapse rate and recovery trajectory. The overall mean of 4.63 (Table 4) demonstrates strong agreement that family involvement significantly and positively influences both the reduction of relapse and the acceleration of overall recovery. The highest agreement was recorded for the statement that 'Family support provides a stable environment that speeds up overall recovery' (mean = 4.78), while high scores were also noted for caregivers' role in recognising relapse warning signs early (mean = 4.58).

Table 4: Influence of Family Involvement on Relapse Rate and Overall Recovery Trajectory (n = 135)

Item	Mean	SD	Level of Agreement
My involvement helps the patient recognize relapse warning signs early.	4.58	0.65	Strongly Agree
I notice fewer relapses when I'm consistently involved in care.	4.53	0.70	Strongly Agree
Family support provides a stable environment that speeds up overall recovery.	4.78	0.50	Strongly Agree
The patient's social functioning improves significantly due to family engagement.	4.64	0.62	Strongly Agree
Overall Mean & SD	4.63	0.61	Strongly Agree

Source: Field Survey Data, UPTH, 2025.

4.5 Barriers to Family Involvement in Patient Treatment

Research Question 4 explored the barriers to optimal family involvement. The overall mean score of 4.31 (Table 5) indicates that family caregivers collectively perceive these factors as significant barriers to their involvement. The two most overwhelming barriers are 'Financial constraints due to high cost of medication/therapy' (mean = 4.87) and 'Caregiver burden (emotional and physical exhaustion)' (mean = 4.71); 'Stigma and fear of discrimination in the community' also scored high (mean = 4.38), suggesting social barriers are pervasive. The lowest-scoring barrier was 'Lack of adequate knowledge about the patient's illness and care needs' (mean = 3.55), still rated moderately, suggesting caregivers are relatively better informed than they are resourced or emotionally supported.

Table 5: Barriers to Family Involvement in Patient Treatment (n = 135)

Item	Mean	SD	Level of Agreement
Stigma and fear of discrimination in the community.	4.38	0.86	Agree
Lack of adequate knowledge about the patient's illness and care needs.	3.55	1.10	Moderate
Financial constraints due to high cost of medication/therapy.	4.87	0.38	Strongly Agree
Hospital policies/visiting hours limit my time with the patient.	4.04	0.87	Agree
Caregiver burden (emotional and physical exhaustion).	4.71	0.55	Strongly Agree
Overall Mean & SD	4.31	0.75	Agree (Significant Barriers)

Source: Field Survey Data, UPTH, 2025.

4.6 Test of Hypothesis

The study tested the null hypothesis that there is no significant relationship between family involvement and treatment adherence among patients with mental illness in UPTH, using Multiple Linear Regression at a 0.05 level of significance to determine the predictive strength of family involvement on treatment adherence (Table 6).

Table 6: Regression Summary – Family Involvement and Treatment Adherence (n = 135)

Model	R	R-Square	Adjusted R-Square	F-statistic	Sig. (p-value)
Family Involvement → Treatment Adherence	0.803	0.645	0.637	78.41	0.000

The regression analysis yielded a significance value of $p = 0.000$. Since $p < 0.05$, the null hypothesis was rejected. The empirical data provide overwhelming statistical evidence to conclude that there is a highly significant positive relationship between family involvement and patient treatment adherence.

5. DISCUSSION

This section interprets the findings in relation to Family Systems Theory (Bowen, cited in Goldenberg & Goldenberg, 2020), which conceptualises families as emotional units in which a patient's recovery is shaped by the family's emotional climate, communication style, and support roles, and existing empirical and theoretical literature on mental health care in low- and middle-income countries, particularly Nigeria.

The finding that family involvement is at a High Extent (overall mean = 4.43) strongly aligns with the sociological reality of healthcare in many sub-Saharan African nations. In a setting like UPTH, characterised by systemic resource limitations and inadequate governmental funding for chronic illnesses like mental health disorders, the family inevitably steps up as the de

facto primary healthcare system. The overwhelming mean score for financial support (mean = 4.84) confirms the family's role as the insurer of last resort, bearing the direct costs of medications, hospital visits, and necessary nutrition, consistent with reports of catastrophic out-of-pocket expenditure borne by households for mental health care in Nigeria (Gureje *et al*., 2020). Similarly, the high scores for medication monitoring (mean = 4.69) and emotional support (mean = 4.75) validate the family as the primary site of care and rehabilitation. However, the lowest score for engaging in treatment-plan decisions (mean = 3.51) reveals a structural contradiction: the family provides the vast majority of financial and logistical input, yet is only moderately included in formal clinical decision-making, suggesting a continued paternalistic model of care in which the caregiver's day-to-day knowledge is often siloed from the professional medical team.

The definitive finding that family involvement has a Very Strong Positive Influence on treatment adherence (overall mean = 4.67) is a cornerstone of this study, empirically validating the central hypothesis. That emotional support (mean = 4.89) was the most potent factor in promoting the patient's willingness to follow care instructions suggests that adherence in mental illness is not a purely logistical challenge but is heavily influenced by the patient's motivation, insight, and emotional state; the family's provision of empathy and a low-Expressed-Emotion environment acts as a buffer against apathy and resistance to treatment. This is consistent with evidence that patients with strong family support are more likely to adhere to treatment regimens (Ogunsemi *et al*., 2021) and less likely to relapse when family participation is coordinated with clinical care (Abiodun *et al*., 2022).

The finding of a Very Strong Positive Influence on reducing relapse and improving recovery (overall mean = 4.63) underscores the family's role in the long-term management of mental illness. The highest agreement, concerning the provision of a stable environment that speeds recovery (mean = 4.78), speaks to the core tenets of family psychoeducation and intervention models, in which therapeutic stability at home reduces environmental stress and hostility, a known predictor of relapse (Esterwood & Saeed, 2021; Zhai & Fang, 2024). Caregivers' strong self-assessment of their ability to recognise relapse warning signs early (mean = 4.58) validates their role as frontline relapse-surveillance agents, enabling timely outpatient intervention that is more cost-effective than managing acute hospital readmission, particularly important in a high-demand, low-resource setting like UPTH.

While involvement is high, the analysis of barriers (overall mean = 4.31) reveals that this participation is under severe stress. Financial constraints (mean = 4.87), the single greatest barrier, reflect Nigeria's limited health insurance coverage and the long-term cost of psychotropic medication, suggesting that the high involvement observed is precariously vulnerable to economic hardship (Ajuwon & Babatunde, 2021). Caregiver burden (mean = 4.71) reflects the psychological and physical exhaustion resulting from the intensity of care demanded; this is clinically significant because caregiver exhaustion is closely linked to high Expressed Emotion, itself a recognised predictor of patient relapse (Simpson & Jones, 2021). Stigma and fear of discrimination (mean = 4.38) confirms that the social environment remains hostile, leading families to practise self-imposed isolation that further intensifies caregiver burden (Akintola, 2024). In summary, while family capacity for involvement in UPTH is high, the financial and psychosocial costs imposed on caregivers are considerable, creating a fragile system in which the two main barriers threaten the sustainability of the very support structure the patient relies upon.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

The findings of this study demonstrate that family involvement significantly influences the treatment outcomes of patients with mental illness at the University of Port Harcourt Teaching Hospital. The presence of supportive family members enhances treatment adherence, emotional stability, and overall recovery; families who actively participate in therapy sessions, medication management, and follow-up care provide patients with the encouragement and sense of belonging necessary for healing and reintegration into the community. Conversely, limited or absent family participation is associated with poor compliance with treatment plans, social withdrawal, and a higher risk of relapse. The study affirms that the involvement of families in mental health treatment is not only beneficial but essential, and that sustained efforts to foster family-centred care within mental health institutions like UPTH will contribute to more effective management, recovery, and societal acceptance of individuals living with mental illness.

6.2 Recommendations

For Hospital Management (UPTH): Mandate the development and delivery of a structured, evidence-based Family Psycho-education Programme (FPEP) led by psychiatric nurses, covering illness management, communication skills, and stress-coping mechanisms for caregivers; allocate specific resources to mitigate financial barriers, potentially through a subsidised drug scheme or a social work programme linking indigent patients and families with existing community and governmental financial support; and implement an institutional anti-stigma campaign targeting staff, patients, and families.

For Nurses and Health Workers: Integrate standardised tools for assessing family strengths, needs, and caregiver burden into the patient admission and discharge process; actively empower family members by delegating appropriate tasks such as observing for early

signs of relapse and supervising medication, ensuring families feel like respected partners in care; and advocate for family rights, ensuring timely and culturally sensitive information about the patient's condition and treatment plan.

For Government and Policymakers: Increase funding for community-based mental health services to support patients and families closer to home, reducing the burden on the tertiary hospital and making sustained family involvement more feasible; and fund and implement wide-reaching, culturally appropriate public health campaigns aimed at reducing the pervasive stigma associated with mental illness across Rivers State and Nigeria.

6.3 Limitations of the Study

The study employed a descriptive cross-sectional design, meaning data were collected at a single point in time; this limits the ability to conclusively determine a causal link between family involvement and future relapse rates, and establishing long-term causality requires a longitudinal study. Since the topic touches on sensitive issues such as stigma and family commitment, respondents may have provided socially desirable responses (e.g., over-reporting involvement or adherence), potentially leading to an upward bias in the observed positive influence. The study was also confined to a single tertiary institution and its specific patient population, and the unique cultural, socio-economic, and institutional context of Port Harcourt may limit the generalisability of the findings to other hospitals or regions of Nigeria.

6.4 Suggestions for Further Studies

Future research should consider a longitudinal design following a cohort of patients over 12–24 months to establish the causal link between different types of family involvement and long-term relapse and recovery outcomes; interventional research evaluating the effectiveness of a structured, nurse-led Family Psycho-education Programme on adherence, relapse, and caregiver burden; qualitative exploration of the lived experience of patients and family caregivers regarding cultural beliefs, stigma, and treatment adherence; and comparative studies examining family involvement and treatment outcomes across different levels of care.

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